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SCIENCE FICTION

JUNE-JULY
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TIME IN THY FLIGHT

by Ray Bradbury

THE OTHER TIGER

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THE FIRE AND THE FLESH

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*** START OF THE PROJECT GUTENBERG EBOOK WAR IN
HEAVEN ***

war in heaven

By Fletcher Pratt

Who is going to use death-rays
when solid-shot and space-mines
will bring far cheaper victory?
An article about space-warfare.

*Virtually all science fiction authors
and most of those who read the stuff
have speculated at one time or
another upon space-warfare. How
will it be waged? What will the ships
be like, the weapons, the tactics?
Fletcher Pratt, even more renowned
as a military and naval historian and
student than as a science fiction
author here take a peek into an all-
too probable future and comes up
with some well-thought-out answers
that are guaranteed to surprise all
would-be spacemen.*

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The mighty ship released a flashing sheet of energy but the Uranian space-vessel's beams were met by a counter-energy screen that caused its blinding heat to ricochet in flashing showers from a barrier invisible against the star-studded black wall of space....

Oh, yeah? It reads well—or has a few thousand fictional times—but it doesn't make sense in a too-real future.

Look chum, a searing bolt of flame has to have something that will support combustion or it will go out. And what do you mean "sheet of energy?" How do you generate it? How do you expect to keep radiation in a tight beam across a couple of thousand miles of space when you can't even prevent a beam of light from spreading after a couple of thousand feet?

The tractor and repulsor beams, screens of force and death-rays of high-power interplanetary stories simply aren't going to work. At least not according to any science we know now. About the only kind of "ray" that might be dangerous would be ultra-violet.

But the Sun itself produces ultra-violet faster and in greater quantity than any generator man could build and unless the crews of space-ships are very thoroughly protected against it there won't be any space-ships.

Maybe the BEMS from Arcturus will come around with something more serious but it is a better-than-even-money bet that they can't do any more about the laws governing radiation than we can. And if they can't space war will have to be fought with far more mundane weapons.

Well, what kind of weapons? That depends upon the kind of ships. You don't expect an earthly battleship to carry torpedoes because she will never get close enough to the enemy to fire them—or a destroyer to mount 16-inch guns because she can't carry them. So the problem of space-war begins with that of designing a space-warship. And that brings up some interesting questions.

The first of them is the shape of the ship. The torpedo-shape with brief wings, the type usually pictured, is very attractive. It is the only shape that could take off from earth and go out through the atmosphere. It keeps the

machinery well away from the living spaces. For landing on the moon or an asteroid it is quite all right because it can be turned over and set down on its tail jets.

But when space-warfare really gets going the torpedo-shape will take a back space before a vessel built in space (probably at a satellite station) to operate and fight in space. And the shape of that ship will be a sphere. It is the strongest, the most economical for the use of the contained cubic capacity, but these are not the main reasons for building space-warships round. The reasons are those of military efficiency, which take precedence over all others when it comes to designing fighting equipment.

In the first place a sphere can be given more than one rocket exhaust. With more than one the spherical ship would have a maneuvering ability making immeasurably superior to the long, graceful torpedo. The latter would have to sweep around in curves of hundreds or thousands of miles, or change its course on gyros, which would take nearly as long. But the sphere, with a simple opposite-direction blast from its rockets, could halt, change course and be off.

Probably only two exhausts are necessary, but those we have to have. I am aware that the engineering problems of building a space-ship that way are very severe but so are the engineering problems of building an atomic submarine or a carrier to carry jet planes. However, when there is urgent military necessity for something, neither expense nor the difficulty of the problem is ever really allowed to stand in the way.

In the second place a sphere can be built with no blind angles of approach. The fighter airplane of today, with an enemy on his tail, is in trouble—and so will be a space-ship with an enemy on its exhaust. Of course, turrets can be mounted above or below the exhaust on a torpedo-shaped ship but they will never give quite the same protection as not having any blind angles at all.

In the third place the sphere is the most convenient shape for landing on the Moon or asteroids and they are going to be important as bases. And in the fourth place the armor of the sphere can carry the main structural stresses, making the interior structure light.

Using a spherical shape means size, of course, but so do several other necessary factors and it is impossible to avoid them all. This means that the warships of space will not be divided into classes of battleships, cruisers and destroyers like warships of the ocean. With an exception to be noted later all will be battleships. There is no reason for making them anything else.

Oceanic destroyers gain speed by sacrificing armor but there is no comparable gain in a space-ship. Once the jets are started and the original inertia overcome the heavy ship will travel as fast as the light one because the limiting factor is how much acceleration the crew can stand physically—and it's the same for both.

On earth the design of a ship is a compromise between the demands of armament, protection, speed and cruising radius, with the last the least important. For the space-ship speed will make its demands but they will not have to be satisfied at the expense of the other characteristics. However, cruising radius is something else.

The ruling consideration in the radius of action of an earth-ship is the ability to carry fuel. Stores for the crew were seldom a problem during the war, even though the food did sometimes run down to Spam and those incredible dehydrated potatoes. Ammunition became a problem in only a few cases. But in the space-navy all this will be changed.

Fuel does not look like a particularly serious problem. A given space-ship will be burning lots while she is using it but most of the time she'll be coasting on gained acceleration and will need fuel only for short bursts of maneuver during action. The true limiting factor in the radius of action of a space-warship is the stores for the crew—not food or water but most specifically air.

The problem with water is not supplying it but getting rid of it. For every five pounds of food you eat two pounds of water is the minimum result, exuded in various ways. And water is ridiculously easy to purify by distillation. Food itself can be carried in various concentrated forms but it is impossible to carry reserve air except in oxygen bottles under compression and it is very difficult to get rid of surplus unwanted carbon dioxide.

For stations in space air-purifiers have been suggested, consisting of algae operating in a churned water-tank. This would be all right for a station which has a steady motion in a determined orbit. But the space-battleship in action will be subject to violent gyrations which would do no good to the air-purifying system even if considerations of weight and space made it practical to install such a system in the first place.

Then there is the added danger that a single hit in so vital an installation would put the ship out of action for keeps while a few oxygen bottles blown up would not matter. So the space-battleship will probably have to depend on bottled air, like a pre-snorkel submarine, and the quantity she can carry will determine her radius of action.

This does not mean that she cannot make quite long voyages, since a ship of the dimensions we are contemplating could store quite a lot of air. But it does help to determine the strategy of space-warfare. It will be fundamentally a struggle for bases where more oxygen can be obtained. Not through going down into the atmosphere of Earth or the thin atmosphere of Mars or the questionable one of Venus. It means bases on the Moon and the asteroids.

The Moon and asteroids are made of rocks, on the surface at least, and practically all rocks are loaded with oxygen—47% in the crust of the Earth for example. With the kind of power that will be available by time we get space-ships, it will be a comparatively simple matter to separate these rock materials from their oxygen electrolytically. Carbon dioxide is partly oxygen, of course, but so stubbornly bonded that no ordinary electrolytic process will break it and it has the unpleasant quality of being a gas under electrolytic conditions.

Since the oxygen-producing machinery will be too heavy and bulky to carry aboard the space-ship the job will have to be done at air-refueling stations and these advanced bases will be the key of space-campaigns. Naturally, they will be powerfully fortified against attack from the enemy's space-fleet. Equally naturally they will be logical points of attack in the hope of limiting the enemy's operations by cutting his bases from under him.

Thus the overall strategy of a space-campaign will somewhat resemble that of the Pacific War, with each party trying to destroy or neutralize the enemy's bases while extending his own. If both have bases on the Moon or

Mars there may even be ground campaigns in support of those in the skies. And getting control of one of those erratic asteroids that come within the orbit of Mars will count for ten.

What about armament? In stories of space-warfare that get away from those improbable rays there is usually some kind of hyper-super-duper torpedo, rocket-propelled. But this only demonstrates that the authors of the stories are thinking in Earth-terms instead of space-terms.

It is perfectly plausible to include some torpedoes, probably with atomic warheads, in the armament of a space-ship—something that would knock out an enemy base or destroy a space-battleship at a single blow. But the torpedo will always be what military men call a weapon of opportunity.

They can't be used at all angles—it would be fatally easy for one of them to make a near-miss on the enemy craft and atom-bomb your own Moon base or some part of the Earth you didn't want atom-bombed. Earth's gravitational attraction would pull such a torpedo in from quite a distance out.

There is also the point that on Earth the torpedo is a comparatively short-range weapon, fired from concealment or under conditions where the torpedo-carrier approaches so rapidly it cannot be stopped—by a submarine under-water, by surface ships at night or from behind smoke-screens, by torpedo-planes that attack at ten times the speed of the target-ship.

None of these conditions can be realized in space-warfare. Even if the space-ship were painted black and operating in a planetary shadow (as in some stories) radar would pick it up at a distance at least equaling that from the Earth to the Moon.

And even black coloring would make the space-ship visible as it occluded the practically continuous blanket of stars visible in open space. You can't make a smoke-screen in space—you can make it, but it wouldn't hang and wouldn't be any use against radar if it did. Comparative speeds that would permit the torpedo-plane type of attack are simply unattainable.

Finally naval experience shows that fire is always opened at the greatest possible range where there is any chance of doing damage—and that range is usually maintained in the hope of avoiding hits. Only when one party has

been so badly pounded that defeat seems inevitable is there an effort to close the range for torpedo or gunfire.

In space-war, given high visibility and the fact that there will be neither gravity nor air resistance to slow up shells, fire will be opened at extremely long ranges—hundreds of miles. At this distance, a rocket-torpedo, clearly indicating its presence by its trail of fire, will be quite easy to intercept—by shooting smaller rockets or shells with proximity fuses at it.

The time it will take the torpedo to cover the distance is the major factor. And running in to launch a torpedo from close range will be very difficult because of the low speed differential between the two parties and again because of the distance. So it is probable that most space-actions will be fought out with guns.

I said guns. But a very different type of gun from those most of us are familiar with, because this is another point where Earth-thinking must give way to space-thinking. When the breech-block of a gun aboard one of our cruisers is swung open to receive a new shell there is a brief moment when the gun forms an open connecting tube from the inside of the turret to the outside air.

This will never do aboard a space-ship—there is no air outside and the air from all over the vessel would rush in a tornado to escape into outer space. There will have to be some device for loading the gun in a vacuum or, since this might give trouble in case of a breakdown or a misfire, more likely an automatic tompion to close the mouth of the gun until there is another charge in the breech. This would have the advantage that the rush of air into the gun, evacuated by the previous firing, would automatically clean it of residual explosion products.

But this is not all. Since there is no air to set up resistances to take a projectile out of its proper path, there is no particular reason for rifling the gun. In fact it would be rather better not to. And while shells were mentioned, in connection with beating off rocket-torpedoes and would be very useful for that purpose, there is no reason whatever for employing them against another space-ship or the dome of an enemy base.

The reason for using a shell against an Earthbound ship is that after it gets through the side it explodes there and messes things up for a considerable radius. But a space-ship is vulnerable in a way that no Earth-ship ever is. If it loses its air it's the finish for everybody aboard.

A high-velocity solid-shot, penetrating the side of a space-ship, could tear holes in several compartments and connect them all with the outside, causing disastrously rapid exhaustion of the air. It would be much more damaging than any shell, which would only ruin one or two compartments. For that matter it was discovered during the last war that a solid projectile which penetrated a tank and ricocheted around inside was quite as effective as a shell-burst.

A solid-shot would have more penetrating power because of its greater mass and unwillingness to break up against armor. And there is also the law of military economics that requires you do everything with the least expensive weapon that will accomplish the purpose. A cast-steel bullet with a soft iron head to "grease" its way through armor will be both cheaper and more effective than any type of shell.

As an alternate type of ammunition for use against ships with exceptionally heavy armor, there might be some projectile using the shaped-charge principle. There's nothing wrong with carrying several different types of projectile for the same guns—both ship and field artillery do it right now.

The type and purpose of the projectiles also determine the size of the guns. The reason for using large-caliber guns on Earth are two—to get a bigger bursting charge at the receiving end and to obtain greater range by a larger driving charge in the breech.

But in space a projectile would have infinite range, unless it fell into the gravitational attraction of some body, and the big bursting charge is not required. So the gun need be only just big enough to make the driving charge give it a very high muzzle-velocity.

Considering the question of ammunition stowage and supply, probably the best caliber would be between 3.5 and 6 inches, 90 and 155 millimeters. Liquid propellants give higher muzzle velocities than solid and would have the advantage of making a better seal around the projectile in the breech. They could be pumped in from metered tubes.

After the air-tight doors have been closed the men who serve these guns will go into action in space-suits, breathing the air around them but having the suits fitted with automatic valves that will close down the moment the compartment begins to lose air. Damage-control parties will have special apparatus for patching holes—probably quick-drying viscous plastics with a silicone base, because of their imperviousness to temperature changes.

What about fire controls? The calculations for them will be arduous because of the distances and speeds involved. But Dr. John D. Clark has pointed out that two space-ships engaged in a fight, no matter what courses they are traveling with relation to the Sun or the planets, are in a single plane with relation to each other and so are all the projectiles fired by both of them.

This plane may tilt violently as they maneuver but at the moment of firing the line from gun to target, or predicted position of target, is a perfectly flat and straight line. This is a much easier ballistic problem than in any Earthly firing, where gravity, air resistance and for long ranges even the curvature and rotation of the Earth must be considered.

The guns will certainly have automatic radar controls and a high rate of fire. The only trouble is that the calculations fed into the controls will have to be made at lightning speeds, in micro-seconds—which means electronic computers. And this brings up another difference between space-warfare and the kind we know, though it may extend to war on Earth.

It is possible to make an electronic computer non-functional by overloading it with data or to drive it electronically insane by feeding it undesirable data. Space-warfare is therefore bound to include all sorts of decoy devices—small metal balloons have been suggested—that will register on the enemy's radar as space-ships and set his guns swinging wildly or firing at phantoms.

The only type of ship to which all that has been said does not apply is the space-minelayer. These can be quite small as compared with the battleships, need carry no armor and can use the torpedo-shape for operating from an earth base. Some years ago Malcolm Jameson suggested small iron spheres as mines, strewn along the enemy's path where he would run into them—but the matter is not quite that simple.

If the space-minelayer merely dumped the spheres overboard as an Earthly minelayer does they would become part of a new small gravitational system around the minelayer, and would follow her around like Mary's little lamb.

The mines of space-warfare will have to be provided with some kind of power that will take them into a predetermined orbit around Earth, the Moon or an asteroid after the minelayer has dropped them. Only enough to set up the orbit, mind—after that they could take care of themselves.

Also it is doubtful whether a mere ball of solid iron would smash up the works of an armored space-ship unless she met it at high speed on an absolute collision course. The more normal event would be for the mine to be picked up by the small but perceptible gravitational attraction of the space-ship and travel with it as a satellite. So the mines of space will probably have to be fairly large and contain heavy explosive charges, probably with proximity fuses.

Still the advantages of mining are so great that they will undoubtedly play a large part in space-warfare. Asteroid or Moon bases can be defended by minefields, which would keep an enemy at a distance until he had painstakingly located the pesky objects and shot them up, during which time the ship would be an ideal target for guns on the ground.

And one striking feature about this space minefield must always be remembered, a feature that makes it different from anything encountered on the ocean. The fields will be constantly moving. Fixed in an orbit 25,000 miles above the Earth a space-mine would always remain over the same Earthly spot and there is a similar critical distance for the Moon, Mars, and the asteroids.

But few of the mines will be at exactly that distance. Unless they are they will be tiny satellites, revolving around their primaries at greater or less speeds, constantly on the go. A space-ship won't have to hit them. They will hunt it up.

There is also the fact that mining in war is not merely a defensive tactic. In World War I the German subs were pretty thoroughly mined in by the

barrages across the North Sea and English Channel. In World War II what was left of the Japanese fleet was immobilized in its harbors by mines. Space-mines would serve a similar purpose of closing a planetoid or an area to all access.

They could serve a tactical purpose as well by limiting avenues of approach. Not that they couldn't be eliminated—rocket-torpedo fire would be something quite different and more effective than the same kind of fire against a space-battleship capable of striking back.

But this would take time and time is the one thing no one can afford to waste in a kind of war where everything will move at speeds exceeding those of the planets.

*** END OF THE PROJECT GUTENBERG EBOOK WAR IN HEAVEN

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